Math Phy Problem set

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1 Instructions

This is a free for all type of problem set, Solve as much of the problems as u can. I dont expect everyone to solve this in 3 hrs. submit the solutions as pdf of pen and paper solutions that u worked out. later check out the solutions for other qns.

Try not to use Ai cuz this is for ur practice, quality of solution over the number will be considered

2 Questions

1) Let ϵ_{ijk} be the levi cita connection and δ_{ij} be the kroneker delta, Prove the following relation between these 2 objects

$$\epsilon_{ijk}\,\epsilon_{lmk} = \delta_{il}\,\delta_{jm} - \delta_{im}\,\delta_{jl}. \tag{1}$$

2) Simplify the following expressions

1. $\epsilon_{ijk}\,\epsilon_{klm}\,\epsilon_{mni}$
2. $\delta_{ij}\,\delta_{jk}\,\delta_{kl}\,\delta_{li}$

3) Prove the following

1. $\nabla^2(\phi\psi) = \phi\nabla^2\psi + \psi\nabla^2\phi + 2\,(\nabla\phi\cdot\nabla\psi)$
2. $\nabla\times(\nabla\times\mathbf{u}) = \nabla(\nabla\cdot\mathbf{u}) - \nabla^2\mathbf{u}$

4) Green's identities: Let S be a closed surface enclosing a volume V .

- a) If ϕ and ψ are two scalar fields, show that

  $$\int_V \left(\phi\nabla^2\psi + \nabla\phi\cdot\nabla\psi\right)\,dV = \oint_S \phi\,\frac{\partial\psi}{\partial n}\,dS \tag{2}$$

  where $\partial\phi\,/\partial n$ is the rate of change of ϕ in the direction normal to the surface element dS . This is called Green's first identity.

- b) Use Green's first identity to show that

  $$\int_V \left(\phi\nabla^2\psi - \psi\nabla^2\phi\right)\,dV = \oint_S \left(\phi\frac{\partial\psi}{\partial n} - \psi\frac{\partial\phi}{\partial n}\right)\,dS. \tag{3}$$

5) Check whether the following sets of elements form an LVS (Linear vector space). If they do, find the dimensionality of the LVS.

1. The set of all antisymmetric tensors of rank 2 in three dimensions.
2. The set of all solutions of the differential equation $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = 0$
3. The set of all n $\times$ n hermitian matrices

6) Prove the following considering properties of the delta distribution

$$\int_0^1 dx_1 \cdots \int_0^1 dx_n\, \delta\big(x_n - \sqrt{x_{n-1}}\big)\, \delta\big(x_{n-1} - \sqrt{x_{n-2}}\big) \cdots \delta(x_2 - \sqrt{x_1}) = 1. \tag{4}$$

7) Let $\sigma_1, \sigma_2\ , \sigma_3$ bet the set of Pauli matrices that form an orthogonal basis and are infinitesimal generators of the lie algebra $su(2)$'s irrep in C^2. Prove the following

1. $\sigma_k \sigma_l = \delta_{kl} I + i\, \epsilon_{klm} \sigma_m$
2. Let $J_k = \frac{1}{2}\sigma_k$, Then $[\mathbf{J}_k, \mathbf{J}_l] = i\, \epsilon_{klm}\, \mathbf{J}_m$.
3. Show that, if σ_k is any one of the three Pauli matrices and α is any complex number, then $e^{ia\sigma_k} = (\cos a)\, I + i(\sin a)\, \sigma_k$
4. (a) If $M = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$, find e^M. Hence write down the eigenvalues of e^M.

8) Derive the addition theorem for Hermite polynomials, Using Rodrigues Formula and Generating functions

$$H_n(x+y) = \frac{1}{2^{n/2}} \sum_{k=0}^{n} \binom{n}{k} H_{n-k}(\sqrt{2}\, x)\, H_k(\sqrt{2}\, y). \tag{5}$$

9) Show that, if (m - n) is even, P is the legandre Polynomial

$$\int_0^1 dx\, P_m(x) P_n(x) = \begin{cases} 0 & \text{if } m \neq n, \\ \dfrac{1}{2n+1} & \text{if } m = n. \end{cases} \tag{6}$$

10) Show the following expansion holds true in terms of Legandre Polynomials, Express the Quadrupole Tensor in terms of Legandre Polynomials

$$\frac{1}{|\mathbf{r} - \mathbf{r}'|} = \frac{1}{\left(r^2 - 2rr'\cos\gamma + r'^2\right)^{1/2}} = \frac{1}{r_g} \sum_{n=0}^{\infty} \left(\frac{r_s}{r_g}\right)^n P_n(\cos\gamma). \tag{7}$$

11) Let The spherical Harmonics be defined as

$$Y_{lm}(\theta, \phi) = (-1)^m \sqrt{\frac{2l+1}{4\pi} \frac{(l-m)!}{(l+m)!}}\, P_l^m(\cos\theta) e^{im\phi}, \tag{8}$$

Show that these Functions are orthogonal and satisfy the following orthogonality Condition

$$\int Y_{lm}^*(\Omega) Y_{l'm'}(\Omega)\, d\Omega = \delta_{ll'} \delta_{mm'} \tag{9}$$

12) From the Fourier series expansion of the periodic function given by $f(x) = x^2$ in the fundamental interval (-1, 1), deduce the value of $\zeta(2)$. where $\zeta(z) \doteq \sum_{n=1}^{\infty} \frac{1}{n^z}$. This is the famous Basel Problem.

13) Prove the following, this setup of finitely infinite number of delta functions is called as a dirac comb, this is the most basic model of a lattice

$$\sum_{n=-\infty}^{\infty} \delta(x - nL) = \frac{1}{L} \sum_{n=-\infty}^{\infty} e^{2\pi n i\, x/L} = \frac{1}{L} + \frac{2}{L} \sum_{n=1}^{\infty} \cos\left(\frac{2\pi n x}{L}\right). \tag{10}$$

14) Prove the following, (Use Poisson's summation formula). This shows up in stat phy so check it out

$$S(a) \overset{\text{def.}}{\equiv} \sum_{n=1}^{\infty} \frac{1}{n^2 + a^2} = \frac{\pi}{2a} \left\{ \coth \pi a - \frac{1}{\pi a} \right\}. \tag{11}$$

15) Show the following $\mathcal{F}^2 = 2\pi P$. $\mathcal{F}$ is the Fourier Transform operator and $\mathcal{P}$ is the Parity Operater where $\mathcal{P}\psi(x) = \psi(-x)$

16) Show that Random Walk in Lattice of dimensionality > 2 is not recurrent

17) Solve this integral using contour Integration, this is know as the Dirichlet Integral

$$\int_0^\infty \frac{\sin bx}{x}\, dx = \frac{\pi}{2} \tag{12}$$

18) Let

$$p_n(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_0 \qquad \text{and} \qquad q_{n+1}(z) = b_{n+1} z^{n+1} + b_n z^n + \cdots + b_0 \tag{13}$$

be polynomials of order n and n + 1, respectively. If C is a simple closed contour that encloses all the roots of $q_{n+1}(z) = 0$ once, in the positive sense, show that

$$\oint_C dz\, \frac{p_n(z)}{q_{n+1}(z)} \;=\; \frac{2\pi i\, a_n}{b_{n+1}}. \tag{14}$$

19) The greens function satisfying the Homogeneous Scalar Wave equation satisfies the condition

$$\left(\frac{\partial^2}{\partial t^2} - c^2\nabla^2\right) G(x, t; x', t') = \delta(x - x')\,\delta(t - t') \tag{15}$$

1. Show that translational invariance implies $G(x, t; x', t') = G(x - x', t - t')$.
2. Show the action of Parity operator on the Green's function, is it invariant under Parity transform
3. Prove that the greens function For 3 spatial Dimensions is $G^{(3)}(R, \tau) = \frac{\theta(\tau)\,\delta(\tau - R/c)}{4\pi R}$ assuming casuality